

Joobong “Joey” Bae

Software Engineer | AI Engineering, Data Science & Full-Stack Development

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EDUCATION	BS Computer Science: Data Science, Software Engineering, and Web & Mobile Programming	
	Liberty University	08.2022 ~ 05.2026
	GPA: 3.62	
	<i>Additional coursework in Web Application Programming, Graphic Design, and Communication.</i>	
EXPERIENCE	Research Assistant , Liberty University, Lynchburg, VA	
		03.2024 ~ 05.2026
	- Architected and scaled PyTorch-based Machine Learning models to analyze 10,000+ data points for structural health monitoring - Implemented rigorous troubleshooting and testing protocols (100+ iterations), significantly improving model prediction accuracy - Collaborated with faculty to optimize model performance through code refactoring and advanced feature engineering	
	Digital Ministry Innovation Team Intern , Trans World Radio, Cary, NC	
		05.2025 ~ 08.2025
	- Deployed AI automation workflows using JavaScript and n8n to streamline content creation and script evaluation - Managed complex API integrations with Microsoft Azure, Google, and ElevenLabs, ensuring 99.9% uptime for automated media processing - Conducted technical research and evaluated state-of-the-art AI tools (Veo 3, Flux, RunwayML) to identify optimal solutions for high-volume video generation - Engineered media processing solutions using FFmpeg to programmatically compile AI-generated assets into final products	
	Graphic Designer & Animator , Treestump Studio, Bothell, WA	
		03.2021 ~ 08.2021
	Designed and animated brand identity and motion graphics for a global YouTube series using Adobe Creative Suite.	
PROJECT	Capstone Project: Duplicate Invoice Detection	
		08.2025 ~ 05.2026
	- Led the React-based UI/UX design and front-end development, collaborating with backend developers to create a side-by-side comparison interface for 20,000+ monthly documents.. - Curated and managed a 25GB testing dataset of invoice data to stress-test the system's accuracy across various document types, including redacted and multi-page files. - Designed interactive dashboards visualizing similarity scores and confidence percentages to facilitate user decision-making.	
RESEARCH	Liberty University Research Week 2025	
		03.2025
	Awarded 3rd place for an NLP sentiment analysis project; developed Prophet/LSTM models for CHD forecasting.	
	EMI Conference and Probabilistic Mechanics & Reliability Conference	
		05.2024
	Developed FCNN and LSTM models to predict cantilever beam motion profiles for structural health monitoring.	
SKILLS	Technical	Python (Pandas, NumPy, Scikit-learn), Java, JavaScript, C/C++, SQL (PostgreSQL, MySQL), React, PyTorch, Time-Series Forecasting (Prophet, LSTM), NLP
	Systems & Tools	Git, Docker, n8n, Tableau, FFmpeg, Linux, Adobe Creative Suite, Cinema4D